SEQUENCE 9

Proportions and percentage change

21–29 January 2027

9.1 Proportion and proportion of a proportion

Key point · Definitions. In a nonempty finite set E, the proportion of a subset F is $p = \operatorname{Card}(F)/\operatorname{Card}(E)$; $0 \leq p \leq 1$. Its expression as a percentage is $100p\,\%$. If $G \subseteq F \subseteq E$, $F \neq \varnothing$, and if p_1 is the proportion of F in E and p_2 that of G in F, the proportion of G in E is $p_1 p_2$.

Example 1. A school has 1500 students, of whom 652 are day boarders. Calculate their proportion. In a class, 60% of the students play sport; among them, 40% are boys. What proportion of the class do the boys who play sport represent?

Solution. $652/1500 \approx 0.4347$, that is about 43.47%. For the class, $0.60 \times 0.40 = 0.24$, that is 24%. The second percentage has the sporty students as its reference, not the whole class.

Example 2. In a fictional school of 800 students, 35% play a sport at a club; among them, 45% play a team sport. Calculate the two counts and the final proportion.

Solution. $800 \times 0.35 = 280$ students at a club. $280 \times 0.45 = 126$ in a team sport. The final proportion is $126/800 = 0.1575 = 15.75\%$.

Watch out. Proportion and rate of change do not answer the same question: a proportion compares a part to a whole, a rate compares a change to an initial value.

9.2 Absolute change and rate of change

Key point · Definition. For an initial value $V_I > 0$ and a final value $V_F > 0$, the absolute change is $V_F - V_I$; the rate of change is

$$t = \frac{V_F - V_I}{V_I}.$$

The rate has no unit. It is positive for an increase, negative for a decrease. The reference is always the initial value.

Example 3. A cheese is sold for 2.13 euros instead of 3.55 euros. Calculate the absolute change and the rate.

Solution. Absolute change: $2.13 - 3.55 = -1.42$ euro. Rate: $-1.42/3.55 = -0.40$, that is a decrease of 40%.

Key point · Multiplier. $CM = V_F/V_I = 1+t$, so $V_F = V_I(1+t)$. To recover V_I, divide V_F by $1+t$ (nonzero).

Example 4. A fictional item priced 50000 Ar increases by 12%, then separately consider a 12% discount on the initial price. Give the final prices. A price after a 20% increase is 72000 Ar; find its initial price.

Solution. Increase: $50000 \times 1.12 = 56000$ Ar. Discount: $50000 \times 0.88 = 44000$ Ar. Last question: $V_I = 72000/1.20 = 60000$ Ar.

Fluency · 5–10 min. Expand $(x-3)^2$; solve $1/x = 5$; give the mean of 8; 10; 12.

Answer. $x^2 - 6x + 9$; $x = 1/5$; 10.

9.3 Successive changes

Key point · Property. Successive changes with rates t_1, t_2 have overall multiplier

$$CM_G = (1 + t_1)(1 + t_2), \qquad t_G = (1 + t_1)(1 + t_2) - 1.$$

We multiply the multipliers; we do not simply add the rates.

Example 5. A manufacturer sold 5600 bicycles in 2022, which is 25% more than in 2021. Its sales decrease by 15% in 2023. Determine the sales of 2021 and 2023, then the overall rate between these two years.

Solution. 2021: $5600/1.25 = 4480$ bicycles. 2023: $5600 \times 0.85 = 4760$ bicycles. Overall multiplier $1.25 \times 0.85 = 1.0625$, so an overall increase of 6.25%. Check: $(4760 - 4480)/4480 = 0.0625$.

Example 6. A price increases by 10% then decreases by 10%. Does it return to the initial price?

Solution. $CM_G = 1.10 \times 0.90 = 0.99$. The final price is 99% of the initial price: an overall decrease of 1%. The two rates refer to different reference values.

Example 7. A rate goes from 20% to 25%. Explain the difference between the increase in percentage points and the relative change.

Solution. The gap is 5 percentage points. The relative change is $(0.25 - 0.20)/0.20 = 0.25$, that is an increase of 25%.

9.4 Reverse change

Key point · Property. To return to the initial value after a change of rate $t > -1$, use the multiplier $1/(1+t)$. The reverse rate is

$$t_{\text{reverse}} = \frac{1}{1+t} - 1 = \frac{-t}{1+t}.$$

Example 8. Which rate compensates a 25% increase? Which rate compensates a 20% decrease?

Solution. After a 25% increase, the reverse multiplier is $1/1.25 = 0.8$: a 20% decrease. After a 20% decrease, it is $1/0.8 = 1.25$: a 25% increase.

Example 9. A price undergoes a 30% discount, then a 10% increase. What additional percentage increase restores the initial price?

Solution. The current multiplier is $0.70 \times 1.10 = 0.77$. The required multiplier is $1/0.77 = 100/77$. The rate is $23/77 \approx 0.2987$, that is about 29.87%. Check $0.77 \times (100/77) = 1$.

Teacher note. Have students state the base of the percentage at each step. The reverse-rate formula is justified by a product equal to 1; memorisation alone is fragile.